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Federated learning for intelligent transportation systems: Emerging trends, challenges and future directions

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ABSTRACT

The growing complexity of urban transportation networks poses significant challenges for data privacy, scalability, and real-time decision-making in Intelligent Transportation Systems (ITS). Traditional centralized models struggle to efficiently process the vast amounts of data generated, increasing the risks of data breaches and latency issues. Federated Learning (FL), a decentralized approach to machine learning, addresses these concerns by enabling collaborative model training without sharing raw data, thus preserving privacy and reducing communication overhead. Despite its potential, the application of FL in ITS remains limited. This survey bridges the gap by reviewing FL's role across key ITS domains, including traffic management, autonomous vehicles, and intelligent infrastructure. We critically assess existing FL frameworks for their ability to meet the demands of ITS, focusing on privacy preservation, communication efficiency, and system scalability. Our findings show that FL significantly improves the operational efficiency and security of ITS, particularly when integrated with emerging technologies like edge computing and artificial intelligence. However, challenges such as data heterogeneity and algorithmic convergence persist and require further investigation. We propose future research directions to enhance FL's adaptability and resilience in ITS, outlining the potential for FL to transform urban mobility by making transportation systems more intelligent, secure, and privacy-focused.

1. Introduction

Intelligent Transportation Systems (ITS) are advanced systems that deliver innovative services across various transport management modes. This empowers users with enhanced information, promoting safer, more synchronized, and 'intelligent' utilization of transportation networks [1]. The primary goal of ITS is to improve the efficiency, safety, mobility, and environmental sustainability of transportation networks [2]. ITS applications span domains such as automated vehicle location, emergency vehicle notification systems, collision-avoidance systems, and variable-speed limit systems [3,4]. Each of these applications leverages extensive data from vehicular and infrastructural sensors, contributing to a dynamic, complex data ecosystem. However, traditional ITS faces significant challenges related to data privacy and security, as well as the need for real-time processing capabilities. Traditional centralized data processing models are becoming increasingly inadequate due to scalability limitations and the risk of single points of failure, which could increase the risk of data breaches or loss [5]. Additionally, these systems often suffer from latency issues that can hinder the effectiveness of real-time responses needed in critical transportation scenarios.

Federated Learning (FL) is a machine learning method that trains an algorithm using data from numerous devices or nodes without sharing the data itself [6,7]. This approach offers a promising solution to many data privacy and management challenges. This approach enhances privacy and reduces the bandwidth required to send data to the cloud, which is particularly beneficial in ITS, where real-time decisions are critical. In the context of ITS, FL can be particularly significant in the current era. By utilizing FL, ITS can achieve better scalability in processing the vast amounts of data generated by traffic infrastructure and enhance the privacy-preserving capabilities of its systems. FL enables ITS infrastructure to benefit from shared learning experiences without exposing their data, thus fostering a more cooperative and intelligent transportation environment [8,9].

The role of FL in ITS is becoming increasingly vital as we move towards more connected, autonomous, and intelligent transportation systems. FL addresses the key challenges traditional ITS systems face and opens new pathways to enhance system efficiency, security, and user privacy. The potential of FL in ITS is vast, encompassing applications ranging from predictive vehicle maintenance to dynamic traffic

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light control and personalized real-time routing [10]. However, deploying FL is challenging. Issues such as data source heterogeneity and infrastructure requirements require significant research [11]. Moreover, integrating FL with emerging technologies such as 5G, edge computing, and AI offers a foundation for innovation but adds layers of complexity to implementation [12]. This paper examines the application of FL to ITS. It reviews current research and key advancements and identifies challenges and opportunities. The goal is to provide a detailed overview of FL's benefits in ITS for revolutionizing urban transportation.

1.1. Research questions and contributions

FL applications in ITS pose distinct challenges, even though FL has been widely studied in domains such as IoT, healthcare, and mobile networks. These challenges are driven by real-time operational demands, highly heterogeneous data, dynamic mobility, and strict privacy requirements. Existing surveys often discuss FL in a broad context, with limited focus on its system-level implications and deployment challenges in ITS environments. To address this gap, this survey is guided by the following research questions:

1.1.1. Research questions

- **RQ1:** Which key ITS services can benefit from FL, and how does FL enhance operational efficiency in these services?
- **RQ2:** How do existing FL frameworks compare in addressing the scalability, security, and integration challenges specific to ITS applications?
- **RQ3:** What are the primary technical and operational challenges in deploying FL within ITS, and what design considerations can mitigate these challenges?
- **RQ4:** What open research issues and future directions remain for the effective adoption of FL in ITS?

1.1.2. Contributions

- **Gap-driven analysis of FL frameworks for ITS:** This survey adopts a gap-driven perspective by providing a focused evaluation of FL frameworks and highlighting ITS-specific algorithmic adaptations.
- **Service-oriented analysis of FL in ITS:** This survey systematically organizes and analyzes FL adoption across core ITS services, including traffic management, autonomous driving, smart parking, and mobility services, addressing the lack of service-level structuring in existing reviews.
- **System-level perspective on challenges and design trade-offs:** Beyond algorithm-centric discussions, the paper examines system-level challenges such as communication efficiency, latency constraints, data heterogeneity, and privacy risks that are often underexplored in prior FL-ITS literature.
- **Identification of open research directions and practical insights:** The survey highlights open research challenges and outlines future directions to guide the design and deployment of scalable, secure, and real-time FL-based ITS solutions.

1.2. The survey organization

The structure of this survey paper is precisely organized to offer a comprehensive overview of FL in the context of ITS. The structure is depicted in Fig. 1. Each section is crafted to guide the reader through the foundational concepts, practical applications, current projects, challenges, and future directions of FL in ITS.

1. **Section II - Foundations:** This section details the fundamentals of FL and ITS. It provides a rationale for adopting FL in ITS and explores how this technology can address specific challenges.
2. **Section III - Existing Works on FL in ITS:** Reviews literature and surveys that examine the application of FL in ITS.

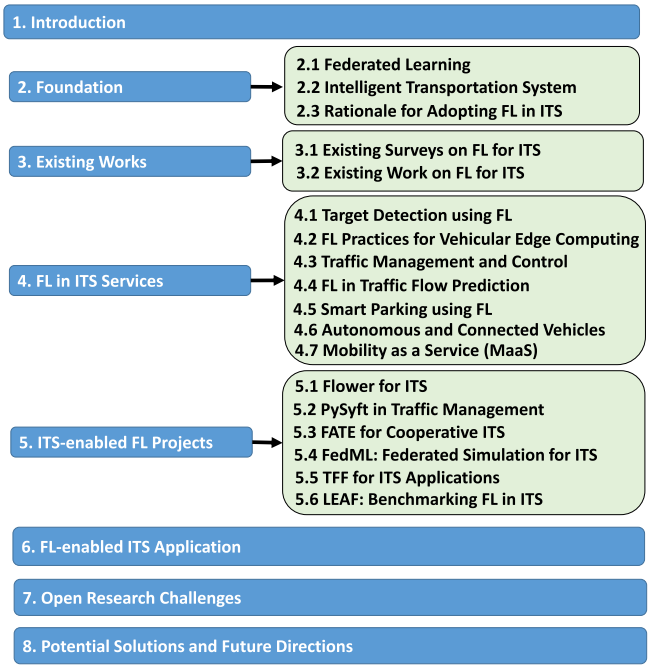


Fig. 1. Survey structure.

3. **Section IV - FL in ITS Services:** Explores the utilization of FL in various ITS services, including traffic management, predictive maintenance, and smart parking.
4. **Section V - ITS-enabled FL Projects:** Showcases practical FL projects and tools such as Flower, PySyft, and FATE, tailored for ITS applications to illustrate real-world viability.
5. **Section VI - FL-enabled ITS Applications:** Delivers insights into specific applications of FL within ITS.
6. **Section VII - Open Research Challenges:** Discusses current challenges faced in integrating FL with ITS, including technical, regulatory, and operational hurdles.
7. **Section VIII - Potential Solutions and Future Directions:** Outlines solutions and future work that reflect the ongoing evolution of FL and ITS and their impacts on future transportation infrastructure.

1.3. Federated learning (FL)

FL is a cutting-edge ML paradigm designed to train algorithms seamlessly across decentralized nodes (servers) without exchanging raw data [13]. This innovative method promotes collaboration between participants or nodes in a machine learning project while protecting their crucial data privacy. Rather than sending sensitive data to a central server, which poses significant privacy risks, FL sends a model to each node for local training. These local model updates are subsequently merged to refine a global model, thus summarizing insights collected from the diverse data held by each participating node. FL operates on the principle of collaborative intelligence, leveraging the collective knowledge stored across distributed nodes while upholding challenging data privacy protocols. This decentralized framework reduces the risks of centralized data storage and enables innovation and scalability in machine learning applications. By empowering devices at the network's edge to actively participate in model training, FL enables real-time learning and adaptation to dynamic data environments, thus marking a new era of privacy-preserving, distributed machine learning methodologies. The workflow of the FL process is also illustrated in Fig. 2.

The central server initiates a learning task, which the client retrieves before configuring their local training environment. The client's timer

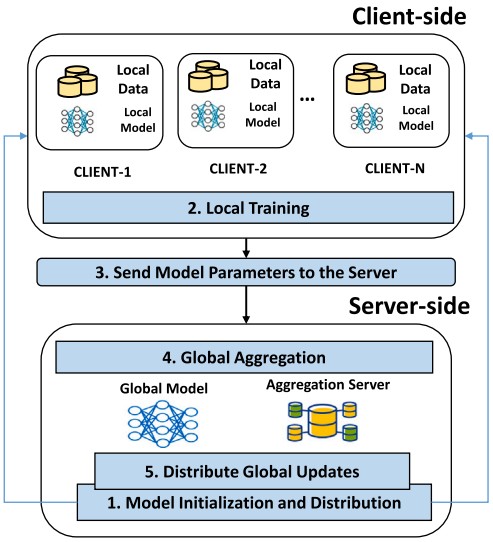


Fig. 2. Federated learning.

starts at the beginning of the second round, and its duration is determined by the average training time achieved by participants in the preceding round. If a client can complete their training within the allocated time, the central server will proceed with the current round's aggregation procedure with their participation. If the client considers that the model they trained for this round will perform worse than the previous round, they may request to forego the aggregation [14]. If the client opts to continue with the aggregation procedure, they notify the primary server to update the model. FL has emerged as a critical solution with the exponential rise in data generation and the growing demand for privacy-preserving technologies. Unlike traditional centralized machine learning models that aggregate all relevant data in a single repository, FL mitigates privacy and security concerns by keeping data decentralized at its source. This approach is crucial for safeguarding sensitive information, such as personal or financial data, from unauthorized access or breaches [15]. Moreover, FL avoids the challenges of transferring large datasets, such as bandwidth constraints and regulatory hurdles, by enabling local model training on distributed devices. FL emerges as a viable choice by complying with local data protection regulations and reducing the need for extensive data transfers, thereby keeping pace with evolving privacy and security standards. Algorithm 1 represents a standard FL workflow and is included as a reference baseline. In practical ITS deployments, such workflows are applied in their original form.

1.4. Centralized vs. federated learning

Traditional centralized ML approaches depend on the continuous transfer of raw data from vehicles, roadside units, and infrastructure sensors to cloud servers [16]. While this design works in controlled or small-scale settings, it often leads to high latency, heavy bandwidth usage, and increased privacy risks. FL offers a practical alternative by allowing models to be trained locally at data sources and limiting communication to model updates only. Vehicular IoT studies have reported that this shift can reduce communication latency by about 21.84% and communication overhead by about 26.32% in federated vehicular network settings [17]. This is mainly achieved by avoiding the transmission of large volumes of sensor data, such as images, video streams, and LiDAR point clouds. At the same time, keeping data local substantially lowers privacy exposure, as sensitive vehicular and user information is no longer collected or stored centrally. These observed advantages explain why FL is increasingly viewed not just as a privacy-preserving solution but as a more suitable learning architecture for scalable, efficient, and real-time ITS deployments. The key differences between centralized and FL with respect to latency, communication overhead, privacy, and scalability are summarized in Table 1.

Algorithm 1 Illustrative Federated Learning Workflow

Require:
K: Total number of clients
N: Number of clients selected in each round
E: Number of local epochs
η: Learning rate
$\{D_i\}_{i=1}^{K}$: Datasets for each client
L: Loss function
Ensure: Updated global model parameters w
Initialize global model parameters w_0 as w_{init}
for $t = 1$ to T **do** ▷ T: Total number of communication rounds
  Select N clients randomly
  for each selected client k **do**
    Initialize local model parameters $w_{t,k} = w_{t-1}$
    for epoch $= 1$ to E **do**
      for each batch in D_k **do**
        Compute gradient: $\nabla L(w_{t,k}, \text{batch})$
        Update local model: $w_{t,k} \leftarrow w_{t,k} - \eta \nabla L(w_{t,k}, \text{batch})$
      end for
    end for
    Compute update $\Delta w_{t,k} = w_{t,k} - w_{t-1}$
    Encrypt and send $\Delta w_{t,k}$ to the server
  end for
  Server aggregates updates: $\Delta w_t = \frac{1}{N} \sum_{k=1}^{N} \Delta w_{t,k}$
  Update global model: $w_t = w_{t-1} + \Delta w_t$
end for
return w_T

Table 1
Comparison of centralized learning and Federated learning.

Metric	Centralized learning	Federated learning
End-to-End Latency	High latency due to raw data transmission and centralized processing that exceed constraints in deployments	Reduced latency (approximately 21.84%) through local training and parallel model updates in FL
Communication Overhead	Very high; continuous transmission of raw sensor data (images, video, LiDAR, GPS) to cloud servers	Low; only model updates exchanged, achieving about 26.32% lower communication overhead
Bandwidth Consumption	Bandwidth-intensive and prone to network congestion, particularly in urban environments	Communication-efficient and scalable to large vehicular networks of more than 100 nodes
Privacy	High risk due to centralized storage and processing of sensitive data	Significantly reduced, as raw data never leaves local devices
Scalability	Limited by centralized server capacity and network bottlenecks	High scalability enabled by distributed computation across edge devices

1.5. Intelligent transportation system (ITS)

ITS involves various wireless and wired communication technologies integrated into transportation infrastructure and vehicles [18]. These systems aim to enhance the safety, efficiency, and sustainability of transportation networks. ITS applications leverage data-driven insights to improve traffic management, reduce congestion, and increase the safety and efficiency of transportation systems. ITS comprises advanced digital and communication technologies for transportation and traffic management systems. These technologies enable integrated applications to enhance the safety, efficiency, and sustainability of transportation networks [19]. ITS uses real-time data collection and processing to

inform decision-making, improve traffic conditions, reduce congestion, and minimize environmental impact. By integrating various technologies, ITS aims to create more intelligent, more connected transportation environments that respond dynamically to real-time conditions and long-term planning needs. ITS can be understood through structured classifications of its components and functionalities [20]. This classification helps in understanding how different technologies and systems contribute to the overall objectives of improving safety, efficiency, and sustainability in transportation networks. The ITS framework is shown in Fig. 3, and the ITS framework categories and subsystems are described in Table 2.

1.6. Rationale for adopting FL in ITS

As ITS evolves, they increasingly handle vast amounts of data generated by diverse transportation modes, sensors, and user devices. Traditional centralized data processing models have struggled with several challenges, making FL an attractive alternative. The integration of FL into ITS addresses many pressing issues and leverages the decentralized nature of transportation data sources. The application of FL in ITS is described in Fig. 4.

- **Communication Efficiency:** Transmitting vast ITS data to a central server leads to bottlenecks and high communication costs [22]. FL reduces this by sending only model updates, which is crucial for real-time data processing in ITS.
- **Handling Data Variety:** ITS data includes structured, unstructured, and semi-structured types [23]. FL simplifies data management by allowing devices to process diverse data locally, improving multimodal data handling.
- **Enhanced Data Analysis:** Centralized systems require complex data fusion, while FL enables efficient integration of diverse data by training a universal model that adapts to various data types, improving analysis efficiency [24].
- **Scalability:** Centralized ITS struggles to scale with increasing data. FL distributes computation across edge devices, reducing strain on central resources and enabling more flexible scalability [25].

1.7. ITS-specific adaptations of FL algorithms

The core structure of FL algorithms is largely domain-agnostic, but their effective use in ITS requires important adaptations. One major challenge is resource heterogeneity, as vehicles, roadside units, and edge servers vary significantly in computational power, energy availability, and communication bandwidth. To address this, ITS-oriented FL implementations often employ adaptive local training schedules and weighted aggregation strategies to prevent resource-constrained clients from degrading overall system performance. Another critical aspect is *mobility-aware client participation*. Unlike static FL settings, ITS environments involve highly mobile clients with transient, unpredictable connectivity. Client selection is often tailored to prioritize nodes with stable connectivity, proximity to roadside infrastructure, or predictable mobility, thereby improving training stability and reducing communication failures. Latency-constrained aggregation is essential for real-time ITS services such as traffic control and autonomous driving. It prioritizes timely model updates over full client participation, enabling faster convergence and improved responsiveness while highlighting the need for ITS-specific optimizations. The mobility and resource-aware client selection workflow (Algorithm 2) and the latency-constrained aggregation mechanism (Algorithm 3) jointly illustrate how FL workflows can be adapted to address mobility, resource heterogeneity, and real-time constraints in ITS environments.

2. Existing work

2.1. Existing surveys on FL for ITS

This section explores various surveys on FL for ITS applications. The surveys examine various aspects of ITS and related technologies, shedding light on their potential contributions and inherent limitations, as depicted in Table 3. Hussain et al. [26] focused on the critical cybersecurity and privacy issues in Connected and Automated Vehicles (CAVs), particularly examining vulnerabilities arising from

Table 2
ITS framework Categories and Subsystems.

Category	Subsystem
Information Systems	Data Acquisition Systems
	Communication Networks
	Data Management and Processing Systems
Control Systems	Traffic Management Systems
	Dynamic Signage Systems
	Emergency Response Systems
Vehicle Systems	On-board Diagnostics (OBD) Systems
	Vehicle-to-Everything (V2X) Technologies
	Autonomous Driving Systems
User Services	Traveler Information Systems
	Ticketing and Payment Systems
	Ride-Sharing and Mobility as a Service (MaaS)
Monitoring Systems	Emission Control Systems
	Weather Monitoring Systems
Integration Systems	Standardization and Protocols
	Systems Integration
Technologies Involved	Cameras, Radar, LIDAR, IoT Devices
Data Analytics and AI	Big Data Analytics, AI Techniques
Data Domain in ITS	Structured, Unstructured, Semi-Structured Data
Storage Infrastructure	Relational, NoSQL, NewSQL Databases
Compute Infrastructure	Batch, Real-Time and Near Real-Time Processing

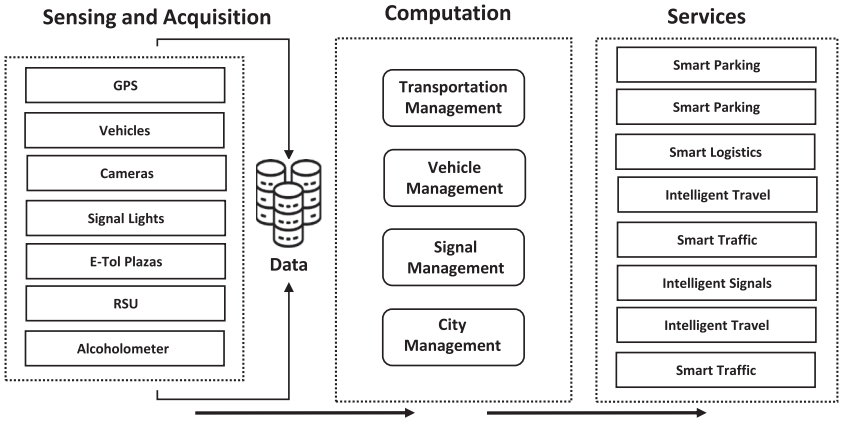


Fig. 3. ITS framework.

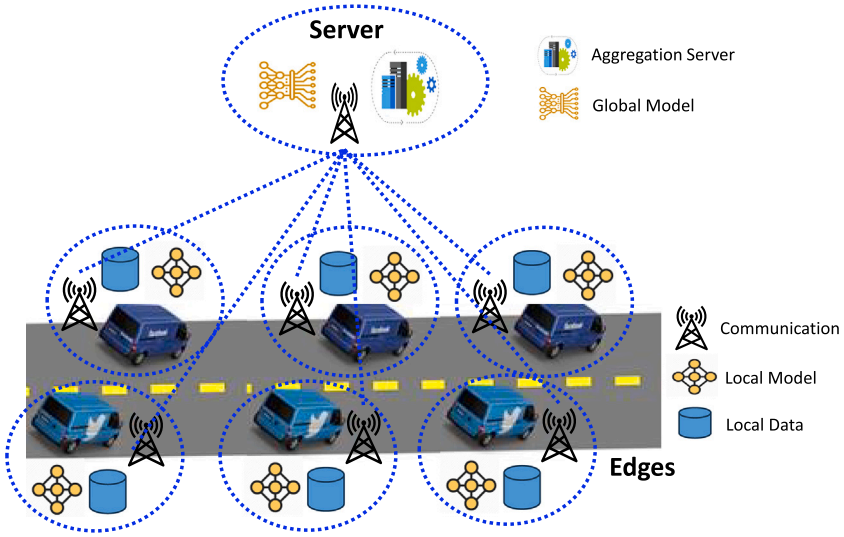


Fig. 4. Application Scenario of FL in ITS.

Algorithm 2 Illustrative Mobility and Resource-Aware Client Selection workflow for ITS-FL.

Require:
C: Set of candidate clients (vehicles/RSUs)
K: Number of clients to be selected
$\hat{\tau}_i(t)$: Connectivity stability estimate for client i
(cpu_i, bw_i, bat_i): Resource profile of client i
$\tau_{\min}$: Minimum stability threshold
$bw_{\min}$: Minimum bandwidth threshold
Ensure: Selected client set S_t
Initialize feasible client set $F \leftarrow \emptyset$
for each client $i \in C$ **do**
  if $\hat{\tau}_i(t) \geq \tau_{\min}$ **and** $bw_i \geq bw_{\min}$ **then**
    $F \leftarrow F \cup \{i\}$
  end if
end for
for each client $i \in F$ **do**
  Compute score:
  $s_i = \alpha \cdot \text{norm}(\hat{\tau}_i(t)) + \beta \cdot \text{norm}(bw_i) + \gamma \cdot \text{norm}(cpu_i) + \delta \cdot \text{norm}(bat_i)$
end for
Select top-K clients based on s_i: $S_t \leftarrow \text{TopK}(F, K)$
return S_t

Algorithm 3 Latency-Constrained Aggregation for Real-Time ITS-FL.

Require:
w_t: Global model at round t
S_t: Selected clients
$T_{\max}$: Latency budget
E_i: Resource-aware local epochs for client i
ω_i: Aggregation weight for client i
Ensure: Updated global model w_{t+1}
Initialize update set $\mathcal{U} \leftarrow \emptyset$ and start latency timer
for each client $i \in S_t$ **in parallel do**
  Initialize local model $w_i = w_t$
  for epoch = 1 to E_i **do**
    Perform local training on client i
  end for
  Compute and send update $\Delta w_i = w_i - w_t$ to server
end for
while elapsed time $\leq T_{\max}$ **do**
  if new update Δw_i received **then**
    $\mathcal{U} \leftarrow \mathcal{U} \cup \{(i, \Delta w_i)\}$
  end if
end while
if $\mathcal{U} = \emptyset$ **then**
  $w_{t+1} \leftarrow w_t$
else
  Aggregate updates:
  $\Delta w = \sum_{(i,\Delta w_i)\in\mathcal{U}} \omega_i \cdot \Delta w_i$
  $w_{t+1} \leftarrow w_t + \Delta w$
end if
return w_{t+1}

attacks in Vehicular Ad-Hoc Networks (VANETs). While their study contributes to understanding these threats, it also highlights limitations, including increased convergence time and reduced accuracy under these attacks, indicating significant challenges in safeguarding VANETs against malicious intrusions. In another venue, Xing et al. [27] examined the intricate structure of the Social Internet of Vehicles (SIoV) and the implications for data security, particularly through the lens of FL. While their work contributes to conceptualizing social communities within vehicular networks, it also highlights the challenges of protecting dynamic, massive datasets within such networks, underscoring the complexities of ensuring data security across evolving SIoV landscapes. Jamil et al. [28] took a broader approach by presenting applications of Decision Trees (DT) and FL in the context of the Internet of Vehicles (IoV). While their study offers valuable insights into potential applications and research directions, it also underscores the challenges of effectively deploying these technologies, including data availability and computing resource constraints, thereby highlighting the practical limitations of implementing DT and FL within IoV frameworks.

Zhang et al. [29] and Zhang et al. [30] provided comprehensive surveys of FL applications in Intelligent ITS, highlighting its potential and the associated challenges. While these studies underscore the benefits of FL over centralized training methods, they also highlight emerging challenges, such as privacy concerns and resource limitations, that may impede its widespread adoption in ITS environments. Tan et al. [31] and Du et al. [32] explored the application of FL in Vehicular Networks (VNs) and the vehicular Internet of Things (IoT), respectively.

Table 3
Existing surveys.

Ref	Contributions	Limitations
[26]	Investigates cybersecurity and privacy concerns in CAVs, focusing on attacks in VANETs.	Attacks increase convergence time and reduce accuracy.
[27]	Studies the SIoV structure, defines social communities by vehicles, and focuses on FL in SIoV data security.	Challenges in protecting massive, dynamic data.
[28]	Presents applications of DT and FL, analyzing surveys and proposing research directions.	Challenges in deploying DT and FL for IoV, including data and computing limitations.
[29]	Surveys FL applications in ITS, addressing challenges and reviewing deployments.	New challenges introduced by FL deployment, including privacy and resource limitations.
[30]	Reviews FL applications in ITS, highlighting benefits over centralized training.	Discusses centralized training limitations like poor real-time performance and data privacy issues.
[31]	Summarizes recent FL applications in VNs, exploring resource management and performance optimization.	Ongoing research challenges related to VNs dynamics and ML distribution needs.
[32]	Surveys FL in vehicular IoT, discussing its significance and technical challenges.	Challenges in utilizing communication, computing, and storage resources efficiently.
[33]	Reviews the application of FL in CAVs, highlighting how FL can improve safety, efficiency, and data privacy by learning from diverse driving environments.	Does not specify limitations directly but highlights challenges and future research needs in applying FL to CAVs.

Table 4
Existing work on FL for ITS.

Ref	Summary	Contributions
[34]	FL for privacy-preserving traffic flow prediction using FedGRU.	Enhances privacy and accuracy without sharing raw data.
[35]	Combines FL and blockchain for cyber-threat detection in ITS.	Ensures safety via blockchain with minimal loss of accuracy.
[36]	Reviews blockchain-enabled FL for Internet of Vehicles (IoVs).	Highlights the security benefits of blockchain with FL.
[37]	Asynchronous FL (DBAFL) for smart public transportation.	Improves FL efficiency and reliability in transport.
[38]	V2X-boosted FL using client selection based on latency.	Enhances FL by optimizing client selection for better performance.
[39]	FL framework for autonomous vehicle controllers.	Tackles non-IID data and mobility with federated proximal algorithm.
[40]	FL in ITS, particularly within 5G networks.	Emphasizes FL advantages in recovery, performance, and privacy.
[41]	Personalized FL with optimal transport for non-IID data.	Introduces FedOT, adapting models to client data distributions.
[42]	CF4FL framework to enhance FL training in transportation.	Reduces convergence time, improving efficiency.
[43]	Explainable FL for taxi travel time prediction.	Improves model transparency and explainability.
[44]	FedSup combines FL with Bayesian techniques for fatigue detection.	Boosts privacy and communication efficiency.
[45]	FL for semantic segmentation in autonomous driving.	Addresses domain shifts and model generalization.
[46]	Federated transfer learning for driver monitoring.	Improves accuracy and reduces communication costs.
[47]	FL for driver activity recognition on edge devices.	Shows strong performance in decentralized edge settings.

While their studies offer insights into resource management and technical challenges, they highlight ongoing research challenges, including dynamics within VNs and the efficient utilization of communication and computing resources within vehicular IoT contexts. Lastly, Chellapandi et al. [33] focused on applying FL to enhance safety, efficiency, and data privacy in CAVs. While their study emphasizes the potential benefits of FL, it also underscores the need for further research to address challenges and future needs in applying FL to CAVs, indicating areas where the technology may not yet be fully mature or optimized for practical deployment. The contributions outlined in these papers offer promising insights into the use of various technologies in connected and autonomous vehicles. However, they also highlight the complex challenges and limitations that must be addressed to maximize their potential in enhancing future transportation systems' safety, efficiency, and security.

2.2. Existing work on FL for ITS

The existing work on FL for ITS and its infrastructure is summarized and provided in Table 4. An FL approach using a Federated Gated Recurrent Unit (FedGRU) algorithm has been proposed to preserve privacy in traffic flow prediction. This method enhances model privacy and prediction accuracy without sharing raw data [34]. Integrating FL with blockchain has been implemented to detect cyber threats in ITS, ensuring data integrity and privacy while slightly reducing accuracy and precision [35]. Blockchain-enabled FL frameworks for the IoV are reviewed, emphasizing the security benefits of integrating blockchain with FL to address privacy and security threats [36]. An asynchronous FL scheme incorporating blockchain and a dynamic scaling factor has been discussed for smart public transportation [37]. A V2X-boosted FL model uses contextual client selection based on communication latency to enhance FL performance by pre-selecting clients based on network conditions [38]. A novel FL framework has been designed for autonomous vehicles, addressing non-IID data distribution and mobility challenges with a dynamic federated proximal algorithm [39].

FL in ITS, especially within the context of 5G networks, has been discussed, highlighting benefits in system recovery, performance, and data privacy [40]. A personalized FL approach based on optimal transport theory effectively handles non-IID data distributions. This method, FedOT, tailors models to each client's data distribution [41]. A communication framework for FL in transportation systems aims to enhance the training process of FL by developing innovative networking components that reduce convergence time and improve system efficiency [42]. Explainable FL for taxi travel time prediction has been explored, proposing methods to enhance model transparency and explainability [43]. FedSup, a framework that combines FL with Bayesian techniques, has been introduced for fatigue detection in IoV scenarios, enhancing privacy and communication efficiency in intelligent systems [44]. Challenges related to domain shifts and model generalization in autonomous driving have been addressed through FL applied to semantic segmentation, proposing solutions for statistical heterogeneity [45]. A federated transfer-ordered-personalized learning framework for driver monitoring focuses on real-world dataset performance, demonstrating significant improvements in accuracy and reduced communication costs [46]. FL-based driver activity recognition for edge devices has been assessed, showing competitive performance in decentralized settings and adaptability to edge computing constraints [47].

Recent studies have increasingly integrated FL with security, privacy, and incentive mechanisms, making these approaches particularly relevant for ITS. Blockchain-based solutions enable secure and anonymous data exchange in vehicular environments [48], while smart contract-based incentives help address fairness and participant motivation in collaborative FL settings [49]. In parallel, recent surveys have examined the adoption of FL in ITS, outlining key use cases, challenges, and opportunities [50]. In addition, cryptographic mechanisms such as encrypted data sharing and lightweight secure aggregation have been proposed to enhance privacy and trust in distributed learning systems [51,52]. Although originally designed for healthcare applications, these mechanisms are transferable to ITS environments, where secure model aggregation and privacy preservation are equally critical. Data integrity auditing mechanisms based on Merkle multi-branch hash trees have been proposed to ensure secure and verifiable cloud storage in next-generation Beyond 5G (B5G) networks, which is particularly relevant for FL-based ITS environments that rely on distributed data storage

and edge–cloud collaboration [53]. Recent advances in deep learning architectures have demonstrated strong capabilities for handling large-scale, heterogeneous data. In particular, hybrid transformer-based models offer improved representation learning and robustness, making them suitable for integration with FL in data-intensive ITS applications [54].

2.3. Existing FL frameworks

FL offers an effective paradigm for addressing privacy, scalability, and data distribution challenges in ITS. Several FL frameworks support the heterogeneous and decentralized nature of ITS deployments, as summarized in Table 5. Flower provides a scalable and framework-agnostic solution for deploying FL across diverse ITS devices [55], while PySyft enables privacy-preserving traffic management through Secure Multi-Party Computation and Differential Privacy [56]. FATE supports large-scale, regulation-compliant collaboration among multiple ITS stakeholders, facilitating secure traffic prediction and monitoring [57]. FedML and TensorFlow Federated (TFF) enable flexible development and decentralized training of FL models on distributed vehicular and infrastructure data [58,59]. LEAF further supports evaluation by providing benchmark datasets and metrics for realistic ITS scenarios [60]. These frameworks highlight the practicality of FL for scalable and privacy-aware ITS applications, but their effectiveness depends on real-world deployment conditions. Flower offers scalability and flexibility with added security efforts; PySyft emphasizes privacy at the cost of greater complexity; and FATE provides robust compliance with higher integration overhead. Together, these tradeoffs show that no single FL framework is optimal for ITS, and practical deployments must balance scalability, security, and ease of integration.

3. Federated learning in ITS services

FL has transformative potential for ITS by leveraging distributed data while maintaining privacy and reducing reliance on centralized data processing. We examine specific ITS services to which FL has been applied, highlighting its impact and citing recent studies.

3.1. Federated learning for target detection in traffic

Image recognition in ITS encompasses tasks such as traffic sign detection and road hazard identification, aiming to improve safety and decision-making [61]. In-vehicle assistance technology enhances driver comfort and safety by detecting emotions and steering angles. However, traffic recognition faces challenges, including limited data availability due to privacy concerns, high resource demands for data transmission, and computational burdens at central processing centers [62]. These issues highlight the need for FL to address efficiency and accuracy in traffic target recognition. Critical challenges in FL for traffic target detection are summarized in Table 6. The algorithmic representation of FL for traffic detection is presented in Algorithm 4. In addition, Fig. 5 illustrates the main FL-based target-detection applications in transportation.

Table 5
Critical Comparison of FL frameworks for ITS.

Framework	Scalability	Security support	Ease of integration in ITS
Flower	Highly scalable with heterogeneous client support	Basic security; relies on external encryption mechanisms	Easy integration; flexible and framework-agnostic
PySyft	Moderate scalability due to privacy overhead	Strong privacy via MPC and differential privacy	Complex integration and higher learning curve
FATE	High scalability for enterprise collaboration	Built-in secure aggregation and compliance support	Moderate integration effort; industrial-grade
FedML	Scalable for experimentation and benchmarking	Supports secure aggregation and privacy extensions	Moderate integration; research-friendly
TFF	Scalable within TensorFlow ecosystems	Limited built-in privacy mechanisms	Restricted flexibility; TensorFlow-dependent

Table 6
Challenges in FL for transportation Target detection.

Challenge	Description
Limited data availability	Many driving images contain personal details, such as the driver's face, making the data unsuitable for sharing.
High resource demand	Transmitting image data requires substantial bandwidth and storage, which can become expensive with large data volumes.
Significant computational load	Centralized data processing centers can experience delays in the recognition process.
The need for rapid processing	Timely decision-making in real-time applications necessitates rapid processing.

3.1.1. Detecting targets on roads through FL

FL has been adopted in road target recognition to address data privacy and real-time data transfer issues. This includes detecting road defects like potholes and traffic signs, which are critical for safety. Alshammari et al. [61] proposed a 3D pothole detection system using You Only Look Once (YOLO) for size estimation. In contrast, Rahman et al. [62] implemented an FL prototype for efficient pothole detection with centralized updates. Yuan et al. [63] introduced adaptive FL to improve model training across diverse edges. Accurate recognition of traffic signs is essential for both manual and autonomous driving. Xie et al. [64] used FL with spiking neural networks to enhance privacy in traffic sign data. Xu et al. [65] applied FL for vehicle target recognition in remote sensing, addressing privacy and training accuracy. Ye et al. [66] developed a selective model aggregation strategy to optimize FL performance across varying computational resources and image quality. Additionally, an asynchronous aggregation protocol improves privacy and training efficiency by selecting models based on local data quality.

3.1.2. In-car detection via FL techniques

FL has greatly improved in-vehicle detection by addressing privacy and computational challenges. For autonomous driving, steering-angle prediction relies on vision-based data, but transferring raw sensor data requires substantial bandwidth. FL minimizes this by enabling vehicles to process data locally and send model updates for global aggregation [67,68]. This reduces communication overhead and improves privacy. Zhang et al. [69] further enhanced this by introducing asynchronous aggregation protocols, thereby improving adaptability and reducing training delays. Driver facial expression detection, critical to safety,

Algorithm 4 FL for Traffic Target Detection.

Require: Data from vehicles D_i for each vehicle i
Ensure: Trained global model parameters w
1: Initialize global model parameters w^0
2: **for** each round $t = 1, 2, \dots, T$ **do**
3: **for** each vehicle $i = 1, 2, \dots, N$ **in parallel do**
4: Train the local model on D_i and update: $w_i^{t+1} \leftarrow w_i^t - \eta \nabla L(w_i^t, D_i)$
5: **end for**
6: Aggregate local models:
7: $w^{t+1} \leftarrow \frac{1}{N} \sum_{i=1}^{N} w_i^{t+1}$
8: Distribute w^{t+1} back to each vehicle
9: **end for**

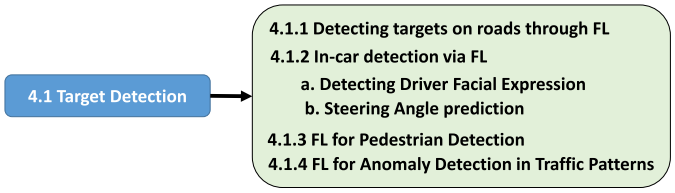


Fig. 5. FL for Target detection in Transportation.

poses privacy risks when data is shared. FL mitigates this by enabling decentralized training for monitoring driver states, such as drowsiness [70,71]. Zhang et al. [72] introduced a transfer learning mechanism and homomorphic encryption to enhance security in drowsiness detection. FL addresses critical challenges in autonomous driving by enabling local data processing, reducing communication costs, and preserving privacy. Promising results have been achieved in steering angle prediction and driver monitoring. Asynchronous aggregation reduces overhead and accelerates training, while homomorphic encryption secures data during aggregation.

$$w^{t+1} = \text{Aggregate_async}\left(\left\{w_i^{t+1}\right\}_{i=1}^{N}\right) \tag{1}$$

$$w^{t+1} = \text{Encrypt}\left(\frac{1}{N}\sum_{i=1}^{N}\text{Decrypt}(w_i^{t+1})\right) \tag{2}$$

3.1.3. Federated learning for pedestrian detection

Enhance pedestrian safety in urban and crosswalk areas by accurately detecting pedestrians under various environmental conditions. Utilizing vision-language semantic self-supervision, vehicles process and analyze images locally, employing advanced semantic recognition to detect pedestrians accurately. This method leverages the context-aware capabilities of pre-trained models to effectively distinguish pedestrians from other objects, thereby enhancing detection reliability across diverse scenarios [73].

- Model Update via Gradient Descent: Each vehicle updates its model parameters locally using gradient descent:

$$\theta_i^{(t+1)} = \theta_i^{(t)} - \eta\nabla L_i(\theta_i^{(t)}) \tag{3}$$

  Where η is the learning rate, and ∇L_i is the gradient of the local loss function concerning the parameters θ.
- Global Model Aggregation: The central server aggregates the model updates from all participating vehicles to form a new global model:

$$\theta^{(t+1)} = \frac{1}{N}\sum_{i=1}^{N}\theta_i^{(t+1)} \tag{4}$$

  Where N is the number of participating vehicles, ensuring the global model benefits from learning diverse pedestrian scenarios across different cars.

3.1.4. FL for anomaly detection in traffic patterns

Identify and respond to unusual traffic patterns and potential hazards to improve road safety and traffic management. Vehicles collaborate to share model updates derived from local data analyses of traffic without exchanging raw data. This collaboration helps recognize and adapt to dynamic traffic conditions by using aggregated insights to detect anomalies, such as unusual vehicle behavior, which may indicate accidents or traffic disruptions [74]. These applications of FL not only bolster traffic management and pedestrian safety but also ensure data privacy and minimize bandwidth usage by processing data locally at the edge. Continuous improvements in model algorithms through FL enhance real-time response capabilities in Intelligent Transportation Systems.

- Local Anomaly Detection Model Training: Each vehicle computes a local anomaly detection model using its collected data:

$$f_i = \text{Detect}_{\theta_i}(D_i) \tag{5}$$

  Where f_i is the anomaly detection function for vehicle i, θ_i are the model parameters, and D_i is the local dataset.
- Local Anomaly Score Calculation: Vehicles compute an anomaly score based on the model's predictions:

$$s_i = \sum_{x\in D_i}\text{score}(f_i(x)) \tag{6}$$

  Where s_i is the anomaly score, and score is a function mapping model outputs to a numerical score indicating anomaly severity.
- Global Anomaly Detection Update: Post aggregation, the global model recalibrates local models:

$$\theta_i^{(t+1)} = \alpha\theta^{(t+1)} + (1-\alpha)\theta_i^{(t)} \tag{7}$$

  where α is a blending factor between the global and local models.

3.2. FL practices for vehicular edge computing (VEC)

This offers an innovative architecture that balances the growing demands for network resources with the massive data needs in automotive environments [75]. By leveraging mobile edge computing, this framework decentralizes computing and storage capabilities, substantially easing network bandwidth constraints and enabling faster service access. Such capabilities underscore VEC as a critical enabler for Intelligent Transportation Systems (ITS) [76,77]. Support for autonomous driving systems is integrated directly at the edge, including On-Board Units (OBUs), Road-side Units (RSUs), and Mobile Edge Computing (MEC) servers. OBUs handle environmental sensing, decision-making, and vehicle control; RSUs provide detailed road and pedestrian data; and MEC servers, positioned near the local devices, aim to reduce system processing loads and data transmission delays, thereby boosting the efficiency of autonomous operations. The role of FL in VEC is depicted in Fig. 6. The FL for VEC is provided in Algorithm 5.

A key application of edge computing in autonomous driving is collaborative perception, which extends sensory range by accessing data from other edge nodes. As autonomous vehicles generate vast amounts of data, immediate local processing becomes essential. Over 60% of data processing will soon occur at the edge. Table 7 outlines the benefits of VEC, including low latency, decentralized computing, high bandwidth, and enhanced privacy. VEC reduces transmission delays, provides fast data access, and enhances security through local processing. Integrating FL with edge computing introduces a strategy in which edge devices train local DL models and only share updates, thereby preserving privacy [32]. This approach supports collaborative learning while maintaining data confidentiality [78]. FL in VEC can be categorized into communication-efficient, resource-optimized, and security-enhanced FL, each addressing specific challenges in this context. Table 8 summarizes FL applications in VEC.

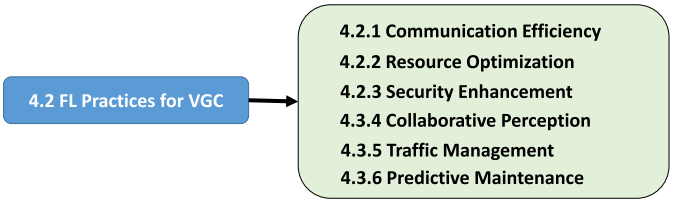


Fig. 6. FL Practices for VEC.

Algorithm 5 Federated Learning in VEC.

Objective: Utilize edge computing to enhance ITS by decentralizing computing and storage capabilities.
Environment Setup:
Deploy OBUs for environmental sensing and vehicle control.
Deploy RSUs to provide detailed road and pedestrian data.
Set up MEC servers near local devices to minimize latency.
Data Handling and Processing:
for each vehicle equipped with sensors **do**
    Collect data from the local environment.
    Process data locally to reduce latency and bandwidth usage.
    Perform real-time analytics to support autonomous driving decisions.
end for
Federated Learning Process:
for each edge device in the network **do**
    Train local DL models using collected data.
    Share model updates, not raw data, with the central server.
end for
Global Model Aggregation:
Aggregate updates at the central server to improve the global model.
Distribute the updated global model back to edge devices.
Outcome:
Enhance autonomous operations' efficiency.
Ensure data privacy and system security.
Optimize resource use across the network.

Table 7
Edge computing benefits for autonomous vehicles.

Benefit	Description
Low latency	Bringing computing resources closer to vehicles minimizes data transmission delays, facilitating immediate responses.
Decentralized computing and storage	This paradigm shifts computing and storage away from centralized locations to edge sites within the vehicle network.
High bandwidth	Offers high-speed access to data and computing resources essential for processing large volumes of data from autonomous vehicles.
Enhanced privacy and security	Local data processing helps safeguard user privacy and increases data security, reducing exposure to breaches and unauthorized access.

3.2.1. Communication efficiency in VEC using FL

Communication overhead between edge devices and central servers is a significant challenge in vehicular contexts with limited network resources [79]. Efficient communication is crucial to managing the data deluge and ensuring the viability of FL. Table 9 describes the subservices of communication efficiency in VEC. Techniques like model compression and selective parameter sharing reduce the amount of data transmitted [80]. Optimization strategies, such as minimizing quantization error and selecting optimal aggregation weights, also reduce communication overhead [81]. Methods such as federated stochastic variance-reduced gradient schemes reduce client–server interactions while maintaining model accuracy, balancing local computations with communication needs [82]. Dynamic aggregation schemes balance synchronous and asynchronous updates, reducing communication costs by allowing flexible participation in the aggregation process [83]. These advancements enable efficient deployment of FL in VEC, allowing it to adapt to network constraints while maintaining scalability and effectiveness.

Table 8
Applications of Federated learning in VEC.

Applications	Key Contributions
Effective communication with FL	Efforts to reduce communication volume, minimize communication rounds, and accelerate model updates to handle data volume in vehicular environments.
Optimized FL	Addressing resource heterogeneity and fault tolerance to minimize training time and improve efficiency.
Security Improved FL	Detection of malicious devices and privacy preservation through blockchain, differential privacy, and homomorphic encryption.
Collaborative Perception	Expands vehicle perception range by obtaining sensor data from other edge nodes, requiring robust, reliable edge computing platforms.
Traffic Management	Utilizes FL for predictive modeling and optimization of traffic flow, enhancing road efficiency and safety.
Predictive Maintenance	Applies FL to analyze vehicle sensor data for early detection of potential faults and proactive maintenance scheduling, reducing downtime and repair costs.

Table 9
Communication efficient improved FL in ITS.

Sub-Service	Description
Reducing Communication Volume	Efforts focus on minimizing data transmission to reduce bandwidth usage, using quantization and selective model transmission.
Reducing Communication Rounds	To boost data exchange efficiency, strategies minimize edge-to-server interactions by leveraging federated stochastic variance-reduced gradient schemes and tailored local training methods.
Accelerating Model Update	Techniques expedite model updates in federated learning by optimizing aggregation methods, using dynamic schemes that adjust server wait times, and using flexible aggregation based on node participation.
Network Bandwidth Optimization	Optimizing network bandwidth by using differential updates, transmitting only model changes instead of the entire model.
Efficient Data Encoding	Utilizing delta encoding or compression to reduce data size while maintaining information integrity.
Dynamic Resource Allocation	Adapting resource allocation dynamically to network conditions and device capabilities to enhance communication efficiency and training performance.
Robust Error Correction	Implementing robust error correction mechanisms to ensure data integrity during transmission and reduce communication overhead.

Minimizing communication overhead

- Reducing Communication Volume: The model compression approach aims to reduce the amount of data transferred between the edge devices and the central server:

$$\theta_{\text{compressed}} = \text{Compress}(\theta, \text{ratio}) \tag{8}$$

where θ represents the model parameters and *ratio* specifies the percentage of parameters to be shared, reducing the total data transmitted.
- Reducing Communication Rounds: Optimizing the number of communication rounds involves a balance between local computations

Table 10
Resource-Optimized FL in ITS.

Sub-Service	Description
Structural Heterogeneity	Algorithms for optimizing resource allocation in VEC account for device diversity. Techniques include min-max optimization, asynchronous FL, and vehicle clustering for parallel tasks.
Fault Tolerance	Strategies manage stragglers by allowing partial computation offloading to edge servers, reducing delays and improving efficiency.
Dynamic Resource Allocation	Adapts resource use dynamically based on network conditions, device capabilities, and task requirements, enhancing FL efficiency in ITS.
Task Offloading Optimization	Optimizes task offloading based on network latency, device load, and energy use, improving resource utilization and reducing training time.
Energy-Efficient Computing	Uses techniques such as DVFS and task scheduling to reduce energy consumption during edge device training, thereby promoting sustainability in FL for ITS.
Data Partitioning Strategies	Develops optimal data partitioning across edge devices that accounts for data size, device capabilities, and communication constraints, thereby improving FL efficiency.

Table 11
Security Enhanced FL.

Sub-Service	Description
Malicious Device Detection	Techniques identify and mitigate malicious edge devices in FL by leveraging blockchain-based incentives, rapid change detection, and collaborative anomaly detection.
Privacy Preservation	Strategies for protecting user data in FL include differential privacy to obscure data and homomorphic encryption for secure computations over encrypted data.
Model Robustness Assurance	Methods enhance the robustness of the FL model against attacks by leveraging adversarial training with adversarial examples and robust aggregation algorithms to prevent model poisoning.
Secure Communication Protocols	Developing secure protocols for data exchange between edge devices and central servers in FL. Using techniques like MPC, encrypted channels, and zero-knowledge proofs for confidentiality and integrity.
Anomaly Detection in Model Updates	Spotting shady model updates from edge devices to prevent malicious behavior or data tampering. Using anomaly detection algorithms and statistical analysis to reject suspicious updates.
Trust Management	Creating trust frameworks for edge devices and central servers in FL. Systems assess device reputation and adjust access based on trust scores to mitigate risks posed by malicious devices.

and the frequency of updates to the central server:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \sum_{i \in S} \nabla L_i(\theta^{(t)}) \tag{9}$$

where η is the learning rate, S is a subset of clients selected based on their data relevance or availability per round, and L_i is the loss for client i.

- Accelerating Model Updates: Dynamic aggregation adjusts the integration of local updates to optimize the aggregation interval:

$$\theta^{(t+1)} = \theta^{(t)} + \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \Delta\theta_i^{(t)} \tag{10}$$

where $\mathcal{A}$ is the set of active clients in a given aggregation window, and $\Delta\theta_i^{(t)}$ is the update from client i.

3.2.2. Resource-optimized FL

FL represents a pivotal approach to collaborative machine learning, enabling edge devices to decentralize model training. However, the heterogeneity in computational capabilities and network conditions across edge devices poses challenges, including imbalanced training durations and inefficient global model aggregation. Weaker devices (stragglers) can significantly prolong the model aggregation phase, while unstable networks can disrupt the training process, thereby diminishing the overall efficacy of FL. To address these challenges, it is crucial to strategically allocate resources and implement fault-tolerance mechanisms to reduce training time and improve FL efficiency. The sub-services of resource-optimized FL in ITS are described in Table 10.

3.2.3. Security enhanced federated learning

The Security Enhanced FL sub-services are described in this section and depicted in Table 11. Detecting malicious devices in FL is crucial, as edge devices may tamper with data, compromising model updates. Techniques like data poisoning detection, outlier detection, and secure aggregation are used to ensure data authenticity and integrity. For instance, Studies applied the quickest change detection method to spot anomalies in model parameters, addressing potential malicious behavior [84,85]. These techniques enhance the trustworthiness of FL by mitigating threats from malicious devices. Additionally, while FL reduces privacy risks by processing data locally, indirect privacy leaks still pose significant challenges, requiring strong privacy-preserving mechanisms.

A key strategy for enhancing privacy in FL is differential privacy, which adds noise to data queries to prevent individual data identification while preserving utility. Homomorphic encryption also secures data exchanges by enabling operations on encrypted data without exposing the data [86]. A combination of blockchain, quick change detection, differential privacy, and homomorphic encryption is used to address security and privacy risks in FL, making FL a robust framework for secure collaborative learning.

- **Malicious Devices Detection:** Anomaly detection can be integrated into the model update process to detect malicious devices:

$$\Delta\theta_i = \begin{cases} \text{Update}(\theta_i) & \text{if Verify}(\theta_i) \\ \text{Reject}() & \text{otherwise} \end{cases} \tag{11}$$

- **Secure Aggregation Technique:** Secure aggregation ensures individual updates remain indistinguishable during aggregation:

$$\theta^{(t+1)} = \text{SecureAggregate}(\{\theta_i^{(t+1)}\}_{i=1}^{N}) \tag{12}$$

- **Differential Privacy:** Privacy is preserved by adding controlled noise to model updates:

$$\theta_i^{(t+1)} = \theta_i^{(t)} - \eta \nabla L_i(\theta_i^{(t)}) + \text{Noise}(\sigma) \tag{13}$$

3.3. Traffic management and control

Traffic management is vital for ensuring the safe and efficient movement of vehicles and pedestrians in urban areas [87]. It uses technologies such as traffic signal control systems and predictive models to regulate flow, minimize congestion, and enhance safety. These systems adjust signals in real time based on sensor data, reducing travel times and emissions, and improving air quality. Efficient management is critical for urban livability, especially as populations grow [88]. FL improves traffic management by enabling local data processing at traffic signals and sensors, allowing more responsive real-time adjustments. FL algorithms support traffic flow prediction based on distributed traffic data, helping

reduce congestion [92]. Devices like cameras and drones analyze traffic data locally and share model parameters rather than raw data [81]. This improves traffic prediction accuracy and supports real-time incident management, pedestrian safety, and dynamic message updates. The integration of FL with IoT further enables smart traffic management, while combining FL with AI promises future advancements in urban mobility. Table 12 describes FL applications in traffic management.

3.4. Autonomous and connected vehicles

FL significantly impacts autonomous driving systems by enhancing data privacy and security while allowing vehicles to learn from collective experiences. In autonomous driving, collision avoidance is a critical safety feature that protects both passengers and pedestrians. By using FL, autonomous vehicles can benefit from shared learning, predictive modeling, and hazard avoidance by leveraging decentralized experiences across numerous vehicles. This decentralized approach enables vehicles to refine their predictive models using real-time data from similar scenarios encountered by other vehicles, thereby enhancing their ability to navigate potential hazards without sharing sensitive data. Another significant application of FL in autonomous vehicles is maintenance prediction. Traditional maintenance scheduling, often based on fixed intervals or reactive responses, can be inefficient. FL improves this by enabling vehicles to analyze operational data across multiple vehicles without exposing individual data, enabling more precise, proactive maintenance scheduling. This reduces downtime and extends the lifespan of vehicle components by predicting maintenance needs in advance [81].

3.5. FL in traffic flow prediction

Research in traffic flow prediction leverages the spatio-temporal dynamics of urban traffic. The Graph Convolutional Neural Network (GCN) is particularly effective in capturing traffic network structures but faces challenges such as data privacy risks, long training times, and high communication costs [89–91]. FL has been introduced into traffic flow prediction to address these. Xia et al. [92] developed an FL-enhanced GCN that balances accuracy with privacy and computational efficiency by leveraging horizontal FL across local road networks, thereby reducing computation while preserving privacy. However, current FL-GNN models often overlook traffic network topologies, exposing them to privacy risks. Zhang et al. [95] addressed this by applying a differential privacy method to adjacency matrices, using Gaussian transformations to protect privacy. Vertical FL (VFL) is also applied to manage diverse spatial data contributions. Yuan et al. [93] proposed a federated graph attention mechanism to capture temporal dynamics while preserving spatial information across regions. Security concerns remain, especially given the vulnerability of central server models to attacks. Blockchain technology has been employed to decentralize the FL update process, thereby improving security [94]. Techniques like local differential privacy and homomorphic encryption further safeguard data. Communication efficiency has also been improved through federated averaging with participant subsampling, reducing communication loads [34]. These advancements highlight FL's potential to enhance traffic prediction performance and security. FL models excel at predicting traffic conditions by leveraging a wide array of decentralized data sources. These models process data locally and forecast traffic patterns, thereby aiding the implementation of preemptive traffic management strategies. The equation for *Traffic Density* is presented as follows:

$$\text{Traffic Density} = \frac{\sum_{i=1}^{N} \text{Vehicle Count}_i}{\text{Area of Region}} \tag{14}$$

Table 12
Applications of FL in traffic management.

Sector	Application of federated learning
Traffic Signal Control Systems	FL processes local data from vehicles and pedestrians, optimizing signals with reduced latency and enhanced privacy.
Traffic Monitoring	Cameras, sensors, and drones use FL to process traffic data locally, minimizing bandwidth and improving privacy.
Data Analytics and Traffic Prediction	FL improves traffic forecasting by leveraging distributed data from cameras and vehicles while protecting privacy.
Dynamic Message Signs	FL updates signs with real-time traffic conditions using collaborative data from multiple sensors.
Incident Management	FL enables faster incident responses by allowing direct communication between vehicles and authorities.
Pedestrian and Cyclist Safety	FL improves safety by optimizing crosswalk signals and light timings based on real-time local data.
Smart Traffic Management	FL with IoT dynamically manages traffic through direct communication between vehicles and traffic devices.
Future Directions	Integrating FL with AI and machine learning will further refine traffic management and support sustainable mobility.

3.6. Smart parking

Smart parking solutions use IoT technology to reduce search times, optimize parking resource management, and improve user experience [95]. Sensors and cameras monitor parking occupancy, with data processed locally or sent to a server to provide real-time information to drivers via apps or signage. These systems often integrate with payment and reservation platforms [96], enhancing convenience and reducing traffic congestion and emissions. Dynamic pricing based on demand helps manage parking during peak times, while data collected can inform better urban infrastructure planning. FL enhances smart parking by enabling decentralized data processing, improving privacy, and reducing bandwidth usage. Each parking sensor processes data locally, training models to predict availability without sharing raw data. FL aggregates local insights into a global model, improving prediction accuracy and efficiency. This approach is particularly beneficial in cities where vast data from numerous sensors must be managed efficiently and privately.

- *Dynamic Pricing Models:* FL is used to adjust parking fees based on real-time data regarding occupancy and demand [95].
- *Parking Spot Prediction:* Systems predict spot availability by learning from local updates, which significantly ease the search for parking and reduce congestion [96].

3.6.1. Federated learning applications in smart parking

FL enhances smart parking by enabling local data processing, allowing sensors and smartphones to detect real-time parking availability without compromising privacy. Table 13 summarizes FL applications in smart parking. FL supports dynamic demand-based pricing, optimizing revenue while ensuring data security. It also provides personalized parking recommendations by analyzing user data on-device, maintaining privacy—predictive maintenance benefits from FL by detecting anomalies locally and reducing data transmission. Additionally, FL improves traffic flow by analyzing movement patterns and occupancy rates and enhances security by processing surveillance footage locally. FL also supports monitoring environmental impacts and sustainable urban practices. FL improves smart parking efficiency, security, and user-friendliness.

3.7. Mobility as a service (MaaS)

FL enables personalized, secure MaaS by integrating diverse transportation data, enhancing route optimization and travel planning without compromising user privacy. The applications of FL in MaaS are

Table 13
Applications of FL in smart parking systems.

Application	Description
Real-time Parking Availability Detection	FL improves parking spot detection by processing data locally on sensors or smartphones, enhancing privacy and prediction accuracy.
Dynamic Pricing Models	FL enables real-time rate adjustments based on demand, optimizing revenue and usage while protecting user privacy.
Optimized Parking Recommendations	FL provides personalized parking recommendations by analyzing local user data, considering preferences, and leveraging usage history.
Predictive Maintenance	Sensors use FL to predict maintenance needs from local data, reducing transmission and improving response times.
Traffic Congestion Management	FL analyzes local movement and occupancy patterns to manage congestion and adjust signals around parking areas.
Security and Surveillance	FL processes surveillance footage locally to detect suspicious activities, ensuring privacy and boosting security.
Environmental Impact Monitoring	FL and environmental sensors monitor local carbon emissions and heat effects, supporting sustainability efforts and urban planning.

Table 14
Applications of FL in MaaS.

Feature	Impact of federated learning
Enhanced Privacy and Personalization	FL processes user data locally on devices or within local environments, enhancing privacy and enabling personalized travel recommendations without exporting sensitive information to central servers.
Real-Time Route Optimization	Utilizes local data from various transportation nodes and user inputs to dynamically update and improve routing algorithms, enhancing travel efficiency and reducing delays.
Predictive Analytics for Demand and Maintenance	FL supports predictive analytics by locally training models on diverse datasets across multiple transport modalities, aiding forecasting of demand surges, potential delays, and maintenance needs.
Intermodal Transportation Integration	Facilitates the seamless integration of different transport modes, enabling various service providers to collaboratively leverage decentralized data to enhance intermodal connections and efficiency.
Secure Data Sharing Across Providers	Ensures secure, private data sharing across service providers in the MaaS ecosystem, supporting collaborative efforts to improve services while maintaining data integrity.
Adaptive Learning for Changing Patterns	FL systems adapt to changing user behaviors and urban dynamics, continuously optimizing MaaS offerings and ensuring responsiveness to user needs and environmental changes.

presented in Table 14. FL models optimize travel routes in real-time by processing data from various transport modes locally [97]. Personalized travel recommendations are generated using FL while respecting users' travel preferences and schedules [98]. MaaS integrates transportation services like public transport, ride-hailing, car-sharing, and bike-sharing through a unified platform for trip planning, booking, ticketing, and payment. FL promotes privacy by processing user data locally, offering tailored travel recommendations without exposing data. It enables real-time route optimization by aggregating local data, improving travel efficiency, and reducing delays. Predictive analytics benefit from FL's local training across diverse datasets, enabling more accurate demand and maintenance predictions without centralized data storage. FL also facilitates seamless integration of various transportation modes, improving intermodal efficiency while maintaining data security. Its adaptability ensures that MaaS systems continuously learn and optimize in response to user needs and urban changes. FL is a pivotal technology for advancing secure, efficient, and user-focused MaaS platforms that transform urban mobility systems.

4. Federated learning projects for intelligent transportation systems

Integrating FL into ITS represents a promising frontier for enhancing transport efficiency, safety, and privacy. This section discusses key FL projects and platforms shaping the future of ITS, as depicted in Table 15.

4.1. Flower for ITS

Flower is an open-source, framework-agnostic FL platform known for its high scalability and support for heterogeneous clients, making it ideal for ITS applications. It facilitates the deployment of federated models across diverse devices and operating systems, which is crucial given the heterogeneous nature of ITS devices such as traffic sensors, in-car systems, and mobile devices used by transport authorities [55].

4.2. PySyft in traffic management

PySyft is a Python library for secure, private deep learning. In ITS, PySyft's integration with popular deep learning frameworks allows for the implementation of privacy-preserving traffic management solutions. Its use of Secure Multi-Party Computation (sMPC) and Differential Privacy ensures that data from sensors and vehicles can be processed without compromising individual privacy, which is a significant advantage in urban traffic systems [56].

Table 15
FL frameworks and their applications in ITS.

Framework	Description and application in ITS
Flower	An open-source, framework-agnostic FL platform ideal for ITS due to its high scalability and support for heterogeneous clients. It facilitates the deployment of a federated model across diverse ITS devices [55].
PySyft	A Python library for secure and private deep learning. Integrated with popular frameworks, it enables privacy-preserving traffic management in ITS through Secure Multi-Party Computation and Differential Privacy [56].
FATE	An industrial-grade FL framework that supports big data collaboration within ITS while complying with data privacy regulations, enhancing traffic prediction and safety monitoring [57].
FedML	Offers a scalable platform for developing and evaluating FL algorithms in ITS, supporting distributed and mobile on-device training [58].
TFF	Extends TensorFlow to support decentralized data processing in ITS, facilitating the development and testing of FL models on data from vehicles and infrastructure [59].
LEAF	A benchmark framework that evaluates FL applications in ITS, providing datasets, metrics, and reference implementations for realistic transportation scenarios [60].

4.3. FATE for cooperative ITS

FATE is an industrial-grade FL framework that supports big-data collaboration while complying with data privacy regulations. In ITS, FATE's modular and scalable architecture enables seamless collaboration among stakeholders, including vehicle manufacturers, transport authorities, and infrastructure providers. This cooperation is crucial for developing comprehensive models that improve traffic prediction, safety monitoring, and vehicle coordination without exposing sensitive data [57].

4.4. FedML: federated simulation for ITS

FedML provides a flexible and scalable platform for developing and evaluating FL algorithms within ITS. Its support for distributed and on-device training makes it particularly suitable for ITS scenarios where

data are not centralized and must be processed locally on devices distributed across the transportation network [58].

4.5. TensorFlow federated (TFF) for ITS applications

TensorFlow Federated (TFF) extends TensorFlow to support decentralized data processing for ITS. TFF enables the development and testing of FL models directly on vehicle and infrastructure data without central aggregation, preserving privacy and reducing bandwidth. The FL and Federated Core layers allow integration of existing ML models for tasks such as predictive maintenance, route optimization, and autonomous driving [59].

4.6. LEAF: benchmarking FL in ITS

LEAF is a benchmark framework for evaluating FL in ITS, offering datasets, metrics, and reference implementations [60]. It allows researchers to assess FL solutions in realistic scenarios, optimizing algorithms for traffic management, vehicle telematics, and environmental monitoring. These frameworks highlight the growing role of FL in ITS, enhancing privacy, scalability, and efficiency for smarter transportation systems.

5. FL enabled ITS applications

ITS aims to improve transportation efficiency, safety, and sustainability. Integrating FL enables the leveraging of distributed data while preserving privacy. This section highlights applications of FL in ITS, summarized in Table 16. FL addresses privacy and scalability challenges, enabling more adaptive and sustainable systems. However, challenges such as algorithmic efficiency and data quality still require attention to fully unlock their potential.

5.1. FL-enhanced traffic management and prediction

One of FL's key applications in ITS is traffic flow management and prediction. Traditional centralized methods often risk privacy and incur high data transfer costs. FL enables local data processing at the network edge (e.g., traffic sensors and cameras), allowing real-time traffic predictions while maintaining user privacy. By analyzing local traffic patterns, FL helps optimize traffic signals, reduce congestion, and improve road safety. Studies such as Zhao et al. [44] have shown that FL supports dynamic model optimization by enabling local models to adapt to specific conditions without exposing sensitive data.

5.2. Autonomous vehicles and FL

Autonomous vehicles (AVs) rely heavily on vast data to navigate safely. FL can play a crucial role in the collaborative learning of AVs by enabling them to learn from decentralized data sources without sharing the actual data. This approach not only helps improve navigation algorithms but also ensures that passenger and route data remain confidential. Research presents a model that uses FL for autonomous driving capabilities through cooperative learning, significantly increasing road safety and efficiency [81].

5.3. FL in public transportation systems

FL is increasingly used to improve public transportation systems by optimizing route planning and schedules using real-time data from various sources, including passenger counts and vehicle locations. This method ensures efficient resource allocation and enhances passenger satisfaction by reducing wait times and overcrowding. A notable implementation used FL to dynamically adjust bus schedules and routes in response to user demand and traffic conditions, demonstrating substantial improvements in service reliability [99].

Table 16
FL applications in ITS.

Application	Description
FL-Enhanced Traffic Management and Prediction	FL enables real-time fatigue detection by processing data locally at edge devices, such as vehicle cameras, supporting dynamic model optimization without compromising privacy [44].
Autonomous Vehicles and FL	FL enhances collaborative learning for autonomous vehicles by enabling them to learn from decentralized data without sharing actual data, thereby improving navigation and ensuring confidentiality [81].
FL in Public Transportation Systems	FL optimizes public transportation by dynamically adjusting routes and schedules based on real-time data, improving resource allocation and passenger satisfaction [99].
Privacy-Preserving Surveillance in ITS	FL supports privacy in surveillance by processing data locally and sharing only model updates, ensuring sensitive video data remains confidential [21].
Smart Parking Systems Using FL	FL uses data from parking sensors and cameras to predict parking availability and optimize management, enhancing privacy by processing data locally [100].
FL for Environmental Monitoring in ITS	FL facilitates anonymous collection and analysis of environmental data from various sensors, supporting real-time monitoring and urban planning while complying with privacy regulations [101].

5.4. Privacy-preserving surveillance in ITS

ITS often requires the deployment of surveillance systems for security and monitoring purposes. FL can help maintain privacy while still enabling effective monitoring. By processing data locally on surveillance devices and sharing only model updates, FL ensures that sensitive video data does not leave the devices. The use of FL for traffic incident detection, in which the system successfully detected and responded to incidents quickly without transmitting raw video data, thereby preserving privacy [21].

5.5. Smart parking using FL

Smart parking systems benefit from FL by using data from parking sensors and cameras to predict parking availability and optimize management. Instead of centralizing sensitive data, FL processes data locally and shares insights through aggregated updates, enhancing user privacy. A study demonstrated that FL can reduce parking search times and optimize space utilization without compromising data security [100].

5.6. FL for environmental monitoring in ITS

Environmental monitoring, such as air quality and noise levels, is essential for sustainable ITS. FL enables the collection and analysis of data from city-wide sensors while preserving the anonymity of individual sources. This supports both real-time monitoring and long-term urban planning. Research demonstrates how FL effectively aggregates environmental data while maintaining privacy compliance [101].

5.7. Experimental case study: FL for vehicle detection in ITS

This section presents a small-scale experimental case study designed to demonstrate representative performance trends of FL in ITS. The objective is to provide empirical insight. The case study focuses on a vehicle detection task using the Udacity self-driving car dataset, which contains approximately 15,000 labeled images distributed across multiple simulated clients to emulate decentralized and heterogeneous ITS environments [102]. Three representative FL algorithms are considered: FedSGD, FedAvg, and FedProx [16]. The results show that FedAvg achieves higher accuracy under balanced data distributions, owing to stable aggregation and reduced communication frequency, as shown

Table 17
Global accuracy Comparison using Balanced data distribution.

FL algorithm	Average global accuracy (%)
FedSGD	78.1
FedAvg	79.0
FedProx	78.6

in Table 17. In contrast, FedProx improves robustness to client heterogeneity by mitigating client drift. FedSGD exhibits lower accuracy and higher communication cost, indicating limited suitability for large-scale ITS deployments. These observations empirically highlight the tradeoffs among accuracy, communication efficiency, and robustness when deploying FL algorithms in practical ITS scenarios.

Figs. 7 and 8 present a lightweight quantitative benchmarking of three representative FL algorithms (FedSGD, FedAvg, and FedProx) over 50 communication rounds under a balanced data distribution. As shown in Fig. 7, FedAvg consistently achieves the highest global accuracy (around 79%), followed closely by FedProx, while FedSGD remains comparatively lower and more variable. These trends are consistent with the summary values reported in Table 17, where FedAvg achieves the highest average accuracy, FedProx remains competitive, and FedSGD trails. In addition, Fig. 8 shows the corresponding global loss behavior: FedAvg generally yields the lowest loss, FedProx remains stable under client variability, and FedSGD exhibits relatively higher loss fluctuations. Together, these results provide a concise empirical illustration of the practical trade-offs between accuracy and optimization stability across common FL strategies in ITS settings. The key benefits of FL in the ITS setting are depicted in Table 18.

5.8. Algorithm complexity and cost-precision tradeoffs in FL-based ITS

FL introduces additional algorithmic complexity compared to centralized learning due to repeated local updates, multiple communication rounds, and coordination among distributed clients. In ITS environments, these factors translate directly into measurable tradeoffs between model precision, communication cost, and computational overhead, as summarized in Table 19. Increas in client participation from low (10%) to high levels (60%) generally improves average model accuracy. It reflects the benefit of incorporating more diverse traffic data [33]. However, this gain comes at the expense of a substantial increase in communication rounds and higher system-wide computation cost. It may not be suitable for latency-sensitive ITS services. Similarly, increasing the number of local training epochs reduces the required number of global rounds and improves accuracy. However, it significantly increases computational load at edge devices such as vehicles and roadside units. Conversely, configurations with reduced client participation or fewer local epochs achieve lower communication and computation costs. However, it may exhibit accuracy degradation, particularly under non-IID and highly dynamic traffic data distributions. Resource heterogeneity further amplifies this tradeoff, as computationally constrained clients may become stragglers, delaying aggregation and affecting convergence. These trends indicate that FL-based ITS deployments must carefully balance accuracy requirements against communication overhead, edge computation capacity, and real-time responsiveness. It makes cost-precision optimization, which is a central design challenge for scalable and reliable ITS systems.

6. Open research challenges

FL has emerged as a transformative technology for ITS, enabling collaborative, privacy-preserving machine learning across distributed networks. Despite its potential, several research challenges must be addressed to fully leverage FL within ITS. This section outlines the open research challenges, as described in Table 20 and illustrated in Fig. 9.

6.1. Communication efficiency

The decentralized nature of FL involves extensive data communication, which can be a bottleneck due to bandwidth limitations and latency in ITS environments. Efficient communication strategies are critical for reducing overhead without compromising learning performance.

6.2. System and data heterogeneity

ITS encompasses a wide range of devices and data sources, from high-end vehicles to simple road sensors, leading to system heterogeneity. Data heterogeneity also poses significant challenges due to the diverse nature of data collected from different geographical locations and contexts.

6.3. Privacy and security concerns

While FL inherently protects data privacy by design, its distributed architecture introduces new security vulnerabilities, such as model poisoning and inference attacks.

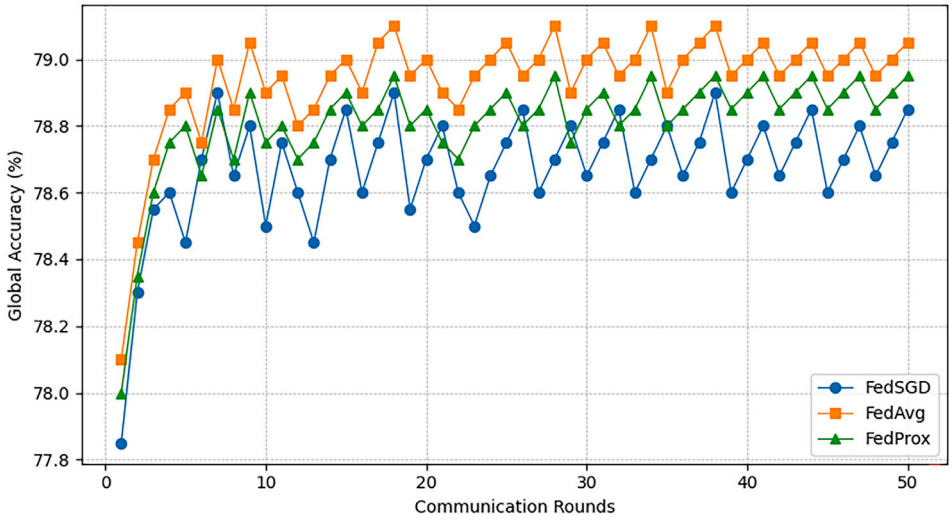


Fig. 7. Global accuracy using FL for Balanced data distribution over 50 communication rounds.

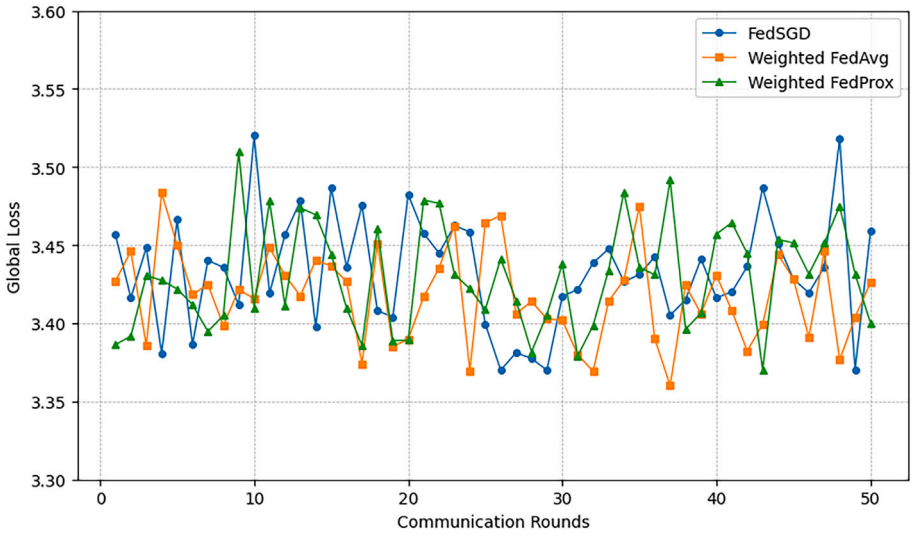


Fig. 8. Global loss using FL for Balanced data distribution over 50 communication rounds.

Table 18
Key benefits of FL in ITS.

FL benefit	Description in ITS context
Privacy Preservation	Raw vehicular data (e.g., images) remains on local devices, reducing the risk of data leakage and ensuring compliance with data protection regulations.
Communication Efficiency	Only model updates or gradients are exchanged during training instead of high-volume raw data, significantly lowering bandwidth usage in congested ITS networks.
Scalability	FL supports training across a large number of distributed clients without central bottlenecks, enabling scalable deployment across city-wide transportation infrastructures.
Data Heterogeneity	FL naturally accommodates non-IID and geographically distributed traffic data, which is common in ITS scenarios due to varying road conditions, driving behaviors, and sensor deployments.
Reduced Latency	Local model training and inference reduce dependence on cloud connectivity, enabling faster response times for latency-sensitive ITS applications such as traffic control and autonomous driving.
Fault Tolerance	Temporary client dropouts caused by mobility or network instability do not halt the learning process, making FL resilient to dynamic ITS environments.
Collaborative Intelligence Across Stakeholders	Multiple entities (e.g., transport authorities, vehicle manufacturers, infrastructure providers) can collaboratively train models without sharing proprietary or sensitive data.

Table 19
Cost–Precision tradeoffs in FL.

FL setting	Avg. accuracy	Comm. rounds	Computation cost
Low client participation (10%)	Lower	Fewer	Low
Moderate client participation (30%)	Moderate	Moderate	Medium
High client participation (60%)	Higher	More	High
Increased local epochs (E=5)	Higher	Fewer	High (Edge)
Reduced local epochs (E=1)	Lower	More	Low (Edge)

Table 20
Challenges of Federated learning in Intelligent transportation systems.

Challenge	Description
Communication Efficiency	The decentralized nature of FL requires efficient communication strategies to manage bandwidth limitations and latency in ITS environments, reducing overhead without compromising learning performance.
System and Data Heterogeneity	ITS includes a diverse range of devices and data sources, resulting in heterogeneous systems and data. This diversity can pose challenges in managing and integrating varied data effectively.
Privacy and Security Concerns	While FL protects data privacy, its distributed architecture may introduce new security vulnerabilities, including model poisoning and inference attacks.
Scalability and Real-time Processing	The need for real-time processing in ITS tasks, such as traffic management, demands scalable FL systems capable of handling large numbers of devices and large data volumes.
Interoperability and Standardization	The lack of standardization and interoperability among ITS components and FL systems can hinder the adoption of FL in transportation.
Ethical and Regulatory Considerations	Deploying FL in ITS entails navigating complex ethical and regulatory issues, including data privacy, decision-making bias, and accountability in autonomous systems.

6.4. Scalability and real-time processing

ITS requires real-time or near-real-time processing for tasks such as traffic management and accident avoidance. To handle the increasing number of ITS devices and data volumes, the scalability of FL systems must be addressed.

6.5. Interoperability and standardization

The lack of standardization across ITS components and interoperability issues among various FL systems can hinder the widespread adoption of FL in transportation systems.



Fig. 9. Research Challenges.

Table 21
Strategies and research directions for Federated learning in ITS.

Challenge	Proposed solutions and research directions
Communication Efficiency	Use model compression techniques like quantization and sparsification to minimize data exchanges; implement asynchronous updates and decentralized communication protocols.
System and Data Heterogeneity	Leverage heterogeneous computing, adaptive learning algorithms, meta-learning, and transfer learning to handle diverse data effectively [103].
Privacy and Security Concerns	Enhance security with anomaly detection, SMPC, differential privacy, and explore new cryptographic techniques to secure FL in ITS [104].
Scalability and Real-time Processing	Utilize edge computing to reduce latency and distribute computational loads; develop lightweight models for faster processing.
Interoperability and Standardization	Establish common standards and protocols; promote collaboration among industry, academia, and government to ensure smooth integration [12].
Ethical and Regulatory Considerations	Create ethical guidelines and regulatory frameworks; engage policymakers, ethicists, and the public to address societal impacts [14].

6.6. Ethical and regulatory considerations

The deployment of FL in ITS must navigate complex ethical and regulatory landscapes, particularly concerning data privacy, bias in decision-making, and accountability in autonomous systems.

As ITS continues to evolve, addressing these challenges through innovative research and development will be crucial to fully realizing the potential of FL to enhance transportation systems. Future research should focus on developing robust, scalable, and secure FL solutions that operate efficiently in the diverse and dynamic environments of ITS.

7. Potential solutions and future directions

As ITS continues to integrate FL, addressing key challenges is essential to advancing the efficiency, security, and practical implementation of these technologies. This section explores potential solutions and strategic directions to enhance communication protocols, accommodate system and data heterogeneity, and ensure robust privacy and security measures, as depicted in Table 21. It also discusses scalability, real-time processing, interoperability, and the ethical dimensions of deploying FL in ITS, paving the way for future innovations and standardized practices.

7.1. Communication efficiency

Developing methods for model compression, such as quantization and sparsification, can significantly reduce the size of data exchanges. Moreover, implementing advanced communication protocols that support asynchronous updates and exploit decentralized topologies may improve scalability and efficiency in ITS applications [22].

7.2. System and data heterogeneity

Adopting heterogeneous computing approaches and developing adaptive learning algorithms that address this diversity is crucial. Techniques such as meta-learning and transfer learning may offer promising solutions for effectively managing heterogeneity [103].

7.3. Privacy and security concerns

Enhancing security protocols, integrating robust anomaly detection systems, and employing secure multi-party computation (SMPC) and differential privacy are potential areas for development. Research into novel cryptographic methods could also bolster the security of FL in ITS [104].

7.4. Scalability and real-time processing

Exploring edge computing frameworks that distribute the computational load and reduce latency can enhance the scalability and responsiveness of FL in ITS. Additionally, developing lightweight learning models that require less computational power could aid in faster processing.

7.5. Interoperability and standardization

Establishing common standards and protocols for FL in ITS is essential. Collaborative efforts among industry, academia, and government to develop unified frameworks could facilitate interoperability and ensure seamless integration of FL technologies [12].

7.6. Ethical and regulatory considerations

Further research is needed to develop ethical guidelines and regulatory frameworks that address these concerns. Engaging with ethicists, policymakers, and the public to understand societal impacts and develop responsible FL applications is crucial [14].

8. Conclusions

This survey has provided an in-depth overview of FL in ITS, emphasizing its role in addressing privacy, efficiency, and scalability challenges in urban transportation. FL enables local data processing across distributed networks, reducing latency and preserving privacy in real-time scenarios. We explored various FL applications, including traffic flow management, predictive maintenance, and dynamic public transportation systems. Ongoing projects demonstrate FL's practical benefits and growing adoption in managing complex transportation systems. Despite these advancements, challenges such as data heterogeneity, model security, and the need for scalable architectures persist. Future research should focus on developing algorithms that efficiently manage diverse, large-scale ITS data and enhance security to counter adversarial attacks. Integrating AI and edge computing with FL will further boost ITS capabilities, making transportation systems more adaptive and responsive to urban demands. In conclusion, FL has the potential to revolutionize ITS by improving efficiency and safeguarding privacy. By addressing current challenges and leveraging emerging technologies, FL can play a key role in developing sustainable, adaptive transportation systems worldwide.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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